NOVA INFORMATION MANAGEMENT SCHOOL

Business Intelligence I

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1. Business Context

The Hollywood film industry operates at the intersection of creativity and commercial strategy. Films are artistic products, but they are also high-risk investments that require careful financial planning. Studios and production companies invest heavily in development, casting, production, marketing, and distribution, often under conditions of significant uncertainty regarding audience reception and financial return. As a result, understanding the factors that drive commercial success has become increasingly critical in an industry characterized by volatility, intense competition, and rapidly changing consumer preferences.

MAD Investment Group is a Lisbon-based fictional company pioneering a new era of film investment. Formed by four investors with complementary backgrounds in data analysis and creative strategy, the team combines analytical intelligence with cultural sensitivity to guide cinematic decision-making. Lisbon, as a growing hub for creative industries and tech innovation, offers strategic access to European and international markets, making it an ideal base for a data-driven film investment firm.

The global film market is complex and highly competitive, with audience preferences shaped by factors such as casting, genre, release timing, and critical reception. Traditional decision-making often relies on experience and intuition and experience, increasing uncertainty and financial risk. To address this challenge, MAD Investment Group applies **Business Intelligence (BI)** principles to film production by analyzing historical box-office data and audience behavior. This approach enables the identification of measurable success patterns and supports investments decisions grounded in evidence rather than intuition, blending creativity with analytical intelligence.

This system allows us to explore what truly drives cinematic success, blending creativity with intelligence. In short, it's our way of bringing data and art together: creativity guided by intelligence, cinema powered by data.

"Truly creative things happen when one thinks differently, yet nobody wants to think differently."

— Steven Soderbergh

2. Business Problems

The core business challenge for MAD Investment Group is the lack of analytical evidence about which creative and market factors most strongly influence a film's commercial and critical performance. Without a structured analytical framework, investment decisions, such as casting, genre selection, or release strategy, are often based on subjective judgment rather than quantifiable insight.

Using Microsoft Fabric, we will ingest, clean, enrich, and model film-industry data to generate actionable insights and support strategic production and investment decisions.

We aim to build a modern data warehouse for cinematic intelligence, a BI-powered decision engine designed to uncover the key factors driving box-office success and audience engagement.

To ensure our Data Warehouse and BI solution directly support our strategic goals, we have identified the key business questions that need to be answered.

BQ1: Which actors appear in the highest-grossing films?

This question identifies which lead actors most frequently participate in films with the highest cumulative box-office revenue. The goal is to reveal associations between actor presence and commercial success, supporting casting decisions and talent valuation.

BQ2: How do award-winning films differ from non-award-winning films in terms of box-office revenue, audience ratings, and genre distribution?

This compares awarded films vs. non-awarded films using financial (revenue) and qualitative (ratings, genre) indicators. It aims to understand whether awarded films tend to have higher audience approval, stronger revenue, or distinct genre tendencies.

BQ3: How does box-office performance differ across runtime segments (<90, 90–120, >120 mins), including ratings?

This evaluates how films of different lengths perform financially and whether runtime is associated with higher/lower ratings. It helps determine whether shorter, standard-length, or long-format films tend to be more successful or better rated.

BQ4: Which genres experience the largest average weekly change in box-office revenue?

This question focuses on weekly revenue variation, identifying which genres grow or decline the fastest over time. It is useful for understanding momentum, decay curves, and genre stability at the box office.

BQ5: How do films perform across filming countries and production-company countries in terms of box-office revenue and audience ratings?

This compares films based on their filming location and the country of their production company. It aims to uncover geographical patterns in film performance, highlighting whether certain countries produce consistently higher-rated or higher-grossing films.

BQ6: What is the average revenue, runtime, and audience ratings (IMDb, Rotten Tomatoes) for films directed by each lead director?

This examines director-specific performance by aggregating revenue, runtime, and ratings by director. It helps identify directors whose films usually attract more viewers, receive higher audience approval, or follow particular production styles.

BQ7: Which genres generate the highest average and total revenues, and how do they differ in ratings, runtime, and release seasons?

This evaluates each genre's financial success (total and average revenue), and compares additional characteristics such as ratings, runtime, and release timing. It provides a multidimensional view of genre performance.

BQ8: How does film revenue differ across rating brackets (IMDb, Rotten Tomatoes)?

By segmenting films into rating ranges, this question shows how audience appreciation (expressed through ratings) is distributed across revenue levels. It identifies whether higher-rated films consistently outperform lower-rated ones or whether revenue behaves differently across rating intervals.

BQ9: How do total and average revenues vary across age classifications, and how does this change across seasons and genres?

This analyzes financial performance by age rating (e.g., PG-13, R). It additionally examines whether age restrictions interact with other factors (seasonality and genre) to influence box-office outcomes.

BQ10: Which weekdays generate the highest average daily revenue and the largest day-to-day revenue change?

This question explores audience behavior throughout the week by identifying which days attract the highest revenue and which shows the greatest fluctuations. It helps us understand consumer behavior patterns such as weekend boosts or weekday drops.

BQ11: Which spoken languages appear most often among the top 50 highest-grossing films?

This question analyzes the linguistic distribution of top-performing films, helping understand which languages dominate among commercial hits and informing decisions about localization, dubbing, or market targeting.

BQ12: Which films rank among the top 5 highest-rated in both high-budget and low-budget categories?

This splits films into high-budget vs. low-budget groups and identifies the top-rated titles in each. It helps compare critical success across different budget levels and understand whether high artistic quality depends on higher production spending.

BQ13: What is the most profitable day, season and marketing season, per year?

This identifies the single most lucrative date, calendar season (winter, spring, summer, fall), and marketing season (e.g., holiday season, award season) for each year. It supports strategic release planning by showing when films historically achieve peak performance.

Together, these business questions form the analytical backbone of our BI initiative, enabling MAD Investment Group to make data-driven film investment decisions, reduce financial risk, and design a repeatable strategy for long-term success in the cinematic market.

3. Data Sources & Collection Strategy

The Business Intelligence solution developed by MAD Investment Group is based on historical box-office performance data, enriched with selected contextual attributes to support analytical decision-making. This section describes the origin, structure, and characteristics of the source data, as well as the data collection approach adopted for the project. The focus is on explaining the nature and limitations of the data rather than providing exhaustive field-level listings, which are documented separately in the appendices.

3.1. Primary Data Source: Box-Office Mojo

Data collection was performed manually, by copying daily box-office tables from the Box Office Mojo website into spreadsheet files. This approach was deliberately chosen to ensure compliance with the platform's usage policies and to avoid automated scraping or excessive requests that could violate ethical or legal constraints.

Each daily table was copied into Excel while preserving the original structure and formatting. This minimized the risk of misalignment or data corruption during ingestion and ensured that the raw data accurately reflected the information published by the source. All data used in the project is publicly available, contains no personal or sensitive information, and is used exclusively for academic and analytical purposes.

3.2. Source Data Structure and Characteristics

At its origin, the source data is organized as a sequence of independent daily snapshots, where each row represents a film's box-office performance on a specific calendar day. Over time, the same film may appear multiple times, reflecting its continued theatrical run.

From an analytical perspective, this structure presents several challenges. Film titles may appear with minor variations, numeric fields may use inconsistent formatting, and performance metrics accumulate over time rather than resetting between periods. Furthermore, the daily structure does not directly support analyses that require weekday patterns or precise temporal alignment with holidays and marketing periods.

These characteristics make the raw source data unsuitable for direct analytical use and motivate the need for a staging layer where consolidation, standardization, and temporal restructuring can occur.

After collection, the spreadsheets were imported into Google Colab for consolidation and transformation using Python. The workflow cleaned and standardized titles, dates, and numeric formats before converting the data into a continuous daily timeline. This allowed each film to be represented with a consistent performance trajectory over time, enabling finer-grained analysis and integration with external datasets.

3.3. External Enrichment Data

To enrich and refine the dataset, a three-phase pipeline was implemented:

<u>Phase 1 - Film Identification & Initial Enrichment</u>

Raw box-office entries were transformed into structured film profiles. Using TMDb as the primary API (and OMDb only when required), we appended core attributes such as lead director, lead actor, and audience-based rating indicators (IMDB and Rotten Tomatoes). A caching layer was introduced to prevent unnecessary API calls and ensure reproducibility across sessions.

Phase 2 - Title-Matching & Accuracy Refinement

To ensure precise film alignment, title normalization and fuzzy matching (via Levenshtein distance[1]) were applied with a ±1-year release tolerance window. Possible film matches were then ranked based on popularity and voting data, which significantly reduced mismatches and strengthened the reliability of the enriched film mappings.

Phase 3 - Contextual Data Expansion

Additional enrichment added contextual attributes, including lead director and lead actor birth years, genre, spoken languages, runtime, and production company country. TMDb remained the authoritative metadata source, with OMDb and Wikidata consulted only for missing values. Persistent caching supported scalability and consistent reproducibility throughout the process.

3.4. Integration and Storage

The final enriched dataset was loaded into the Microsoft Fabric Lakehouse, enabling scalable storage, star-schema modeling, and BI-driven analytics. A 34,568-row sample was used for development and testing, while the full dataset remained available for final analytical work.

3.5. Scope and Limitations of the Source Data

While the source data provides a strong foundation for box-office analysis, it also imposes clear limitations. The dataset focuses on domestic theatrical performance and does not capture international box-office results, streaming performance, or post-theatrical revenue streams. In addition, some contextual attributes are incomplete or missing for certain films, reflecting the limitations of publicly available sources.

[1] Levenshtein distance measures how different two strings (texts) are by counting the minimum number of edits needed to turn one into the other.

These constraints were explicitly acknowledged during project design and informed modeling decisions throughout the staging, ETL, and Data Warehouse layers. Rather than attempting to artificially complete or infer missing information, the project prioritizes analytical transparency and fidelity to the underlying data.

4. Staging Area Role

The Staging Area represents a dedicated database layer positioned between the original source data and the analytical Data Warehouse. Its primary role is to store an integrated and governed representation of the source data that remains faithful to the original semantics while being structurally suitable for downstream analytical processing.

In this project, the Staging Area was introduced to isolate source-related data issues, such as heterogeneity, incompleteness, and semantic ambiguity from the analytical layer. By separating data preparation concerns from analytical modeling, the architecture improves traceability and maintainability, while ensuring that analytical results are based on a clearly defined and controlled data foundation.

4.1. Staging Area Configuration

From an architectural perspective, the Staging Area was configured as a separate database environment, physically and logically distinct from the Data Warehouse. This database contains relational tables designed to hold prepared source data without imposing analytical structures such as star schemas, dimensional hierarchies, or performance-oriented optimizations.

The Staging Area schema mirrors the logical structure of the integrated source data, using descriptive attributes and business identifiers rather than surrogate keys. No analytical assumptions are enforced at this level, allowing the Staging Area to function as a neutral and controlled intermediary between raw source data and dimensional modeling.

This configuration ensures that the Staging Area acts as a stable reference point for all downstream processes, while preserving the ability to trace analytical results back to the original data characteristics.

4.2. Data Representation and Structural Alignment

Data stored in the Staging Area follows consistent structural conventions across entities and attributes. Textual attributes are represented using standardized formats to enable reliable comparison and grouping, while numerical and date attributes are stored using harmonized data types to eliminate ambiguity arising from heterogeneous source representations.

The Staging Area does not attempt to enrich or reinterpret the source data. Instead, it provides a structurally aligned representation that supports consistent downstream processing while maintaining fidelity to the original information content. This approach ensures that all subsequent analytical logic is applied to a well-defined and coherent data structure.

4.3. Handling Missing Values and Semantic Ambiguity

A core design principle of the Staging Area is the correct handling of missing and semantically ambiguous values. In several attributes, missing values may represent different real-world situations. For example, a NULL value may indicate either the absence of a characteristic or missing information in the source data.

The Staging Area also serves as the primary layer for enforcing semantic governance. Throughout this layer, data is represented in a transparent and minimally transformed manner, ensuring that limitations and uncertainties present in the source data remain visible and auditable.

This governance-oriented approach supports trustworthy downstream analysis and ensures that analytical results can be interpreted with a clear understanding of the underlying data semantics.

4.4. Governance Configurations

The Staging Area also serves as a key governance layer within the overall architecture. By providing a controlled and standardized representation of the source data, it establishes a single, authoritative reference point for subsequent processing and validation.

Semantic distinctions present in the source data, such as differences between explicitly stated values and missing information are preserved to avoid conflation. Data quality limitations identified at source level are documented rather than silently corrected, ensuring traceability and supporting informed analytical use.

Through this governance-oriented design, the Staging Area supports both data reliability and analytical transparency.

4.5. Relationship with the Data Warehouse

Once data has been integrated, structurally aligned, and governed within the Staging Area, it becomes suitable for loading into the Data Warehouse. At this point, the responsibilities of the Staging Area end, and dimensional modeling and analytical optimization begin.

The Data Warehouse differs fundamentally from the Staging Area in purpose and structure. While the Staging Area focuses on source fidelity and governance, the Data Warehouse is optimized for analytical querying through star schemas, surrogate keys, and performance-oriented design choices.

This clear separation of concerns results in a clean analytical layer built on reliable, well-understood data, and reinforces the robustness of the overall Business Intelligence solution.

5. Dimensional Data Warehouse

The Data Warehouse developed by MAD Investment Group was designed to support evidence-based decision-making in film investment by enabling structured, high-performance analysis of box-office performance across time, content, talent, geography, and audience reception. The design of the warehouse was driven directly by the business needs defined in section 2 and implemented using a dimensional modeling approach suitable for analytical workloads.

5.1. Modeling Approach and Design Objectives

A dimensional modeling methodology based on a star schema was adopted for this project. This approach was selected because it aligns with the analytical nature of the business questions, supports intuitive slicing and aggregation, and provides efficient query performance in a Business Intelligence environment.

The core business process modeled in the Data Warehouse is the daily box-office performance of films. All design decisions regarding facts, dimensions, and relationships were derived from this process and guided by the requirement to answer the defined business needs without introducing unnecessary complexity.

5.2. Star Schema Overview

The Data Warehouse is structured as a star schema centered around a single fact table, fact_daily_box_office, supported by five conformed dimensions: dim_date, dim_film, dim_actor, dim_director, and dim_production_company.

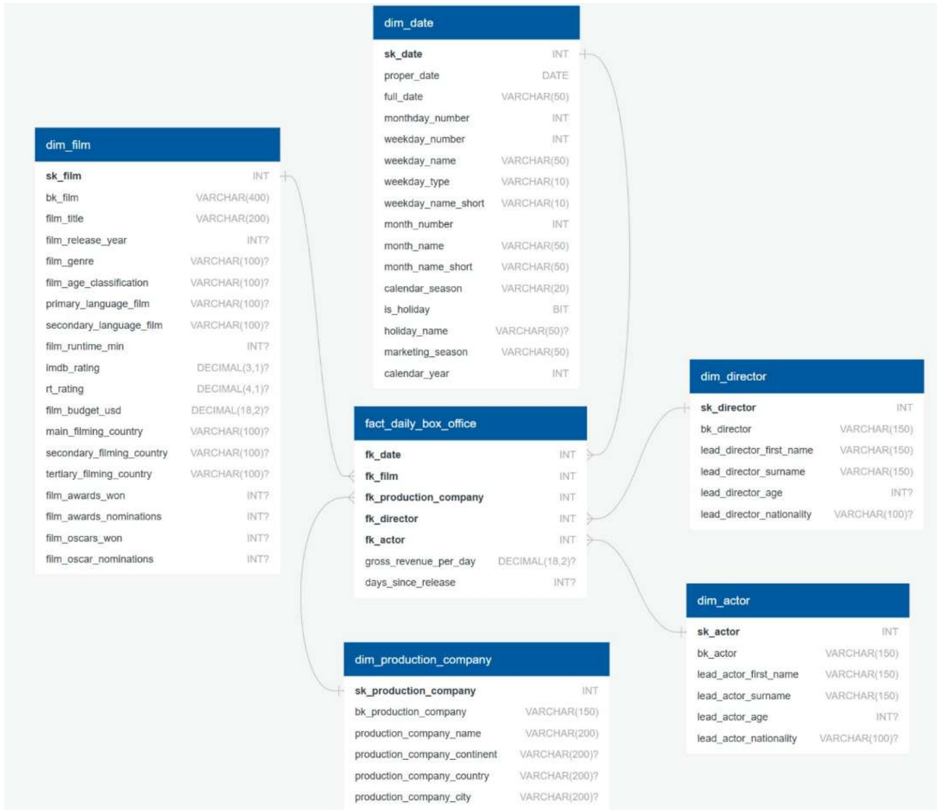


Figure 1: Dimensional (Star Schema) Model of the Daily Box Office Data Warehouse.

This structure allows MAD Investment Group to analyze box-office performance from multiple perspectives while preserving a consistent analytical grain.

5.3. Fact Table Definition and Grain

5.3.1. fact_daily_box_office

The central fact table, **fact_daily_box_office**, captures the commercial performance of films at a daily level.

The grain of the fact table is explicitly defined as:

One record per Film × Date

Defining the grain before selecting facts and dimensions ensured that all analytical requirements could be met without introducing aggregation inconsistencies.

Each row represents the box-office performance of a specific film on a specific calendar day. This grain was chosen because it supports all required analyses, including weekday behavior, seasonal trends, revenue decay, cumulative performance, and aggregation to higher analytical levels such as film, actor, or director.

To contextualize each performance record, the fact table includes foreign keys referencing every dimension:

Attributes	Data Type	Nullable	Description
fk_date	INT	No	Foreign key → dim_date
fk_film	VARCHAR(400)	No	Foreign key → dim_film
fk_production_company	VARCHAR(150)	No	Foreign key → dim_production_company
fk _director	VARCHAR(150)	No	Foreign key → dim_director
fk _actor	VARCHAR(150)	No	Foreign key → dim_actor

Table 1: Foreign Key Structure of the Daily Box Office Fact Table.

These foreign keys ensure that every measure can be interpreted within the full set of descriptive attributes across the dimensions. In the fact table, the primary key is defined implicitly as the composite of all foreign key. This composite key ensures the integrity of the daily film × date grain and guarantees that no two records represent the same film on the same calendar day with the same dimensional context.

The fact table includes the following measures:

- **revenue_per_day**, representing daily box-office revenue
- **days_since_release**, representing the number of days that have passed since the movie's premiere

These measures directly support the evaluation of commercial success, growth dynamics, and long-term performance patterns.

Attributes	Data Type	Nullable	Description
revenue_per_day	DECIMAL(18,2)	Yes	Daily revenue
days_since_release	INT	Yes	Days since the movie's premiere

Table 2: Measures of the Daily Box Office Fact Table.

5.4. Dimension Tables and Business Relevance

5.4.1. dim_film

The film dimension stores descriptive attributes that characterize each movie, as shown below in Table 3. This dimension enables analyses related to genre performance, runtime segmentation, audience reception, award impact and budget comparison.

By concentrating film-level descriptors in a single dimension, the model supports rich analytical comparisons while keeping the fact table focused exclusively on measurable performance.

Attributes	Data Type	Nullable	Description
sk_film	INT	No	Surrogate key for film
bk_film	VARCHAR(400)	No	Business key for film
film_title	VARCHAR(200)	No	Film title
film_release_year	INT	Yes	Year of release
film_genre	VARCHAR(100)	Yes	Genre of the film
film_age_classification	VARCHAR(100)	Yes	Film's age classification
primary_laguage_film	VARCHAR(100)	Yes	Film's primary language
secondary_language_film	VARCHAR(100)	Yes	Film's secondary language
film_runtime_min	INT	Yes	Film runtime in minutes
imdb_rating	DECIMAL(3,1)	Yes	IMDb rating film score
rt_rating	DECIMAL(4,1)	Yes	Rotten Tomatoes film score
film_budget_usd	DECIMAL(18,2)	Yes	Film's production budget in USD
main_filming_country	VARCHAR(100)	Yes	Primary country filmed

secondary_filming_country	VARCHAR(100)	Yes	Secondary country filmed
tertiary_filming_country	VARCHAR(100)	Yes	Tertiary country filmed
film_awards_won	INT	Yes	Film award wins
film_awards_nominations	INT	Yes	Film award overall nominations
film_oscars_won	INT	Yes	Film Oscar wins
film_oscar_nominations	INT	Yes	Film only Oscar nominations

Table 3: Film Dimension Attributes.

5.4.2. dim_date

The date dimension provides comprehensive temporal context, including standard calendar attributes as well as enriched indicators such as weekday type, holidays, calendar seasons, and marketing seasons. Holiday and season definitions were aligned with the classification logic used by the source data rather than restricted to official federal holidays, ensuring analytical consistency with observed audience behavior. This dimension supports analyses related to weekday patterns, seasonal performance, holiday effects, and release timing optimization.

Attributes	**Data Type**	**Nullable**	**Description**
sk_date	INT	No	Surrogate key for the date
proper_date	DATE	No	Actual calendar date
full_date	VARCHAR(50)	No	Full date string
monthday_number	INT	No	Day of the month
weekday_number	INT	No	Day of the week
weekday_name	VARCHAR(50)	No	Name of the weekday
weekday_type	VARCHAR(10)	No	Weekday or Weekend
weekday_name_short	VARCHAR(10)	No	Short name of the weekday
month_number	INT	No	Month number
month_name	VARCHAR(50)	No	Name of the month
month_name_short	VARCHAR(50)	No	Short name of the month
calendar_season	VARCHAR(20)	No	Calendar season

is_holiday	BIT	No	1 if proper date is holiday
holiday_name	VARCHAR(50)	Yes	Holiday name if applicable
marketing_season	VARCHAR(50)	No	Movie-industry marketing season
calendar_year	INT	No	Calendar year

Table 4: Date Dimension Attributes.

<u>Hierarchies:</u>

- calendar_year → month_name → proper_date
- calendar_year → marketing_season → is_holiday → holiday_name
- weekday_type → weekday_name → proper_date

5.4.3. dim_actor & dim_director

The actor and director dimensions store descriptive information about the lead actor and lead director associated with each film. In the dataset, each film is associated with exactly one lead actor and one lead director.

This design choice does not imply that revenue is generated independently by actors or directors. Instead, these dimensions provide contextual slicing, enabling actor- and director-based analysis of daily film performance while preserving a consistent grain.

Attributes	Data Type	Nullable	Description
sk_actor	INT	No	Surrogate key for actor
bk_actor	VARCHAR(150)	No	Business key for actor
lead_actor _first_name	VARCHAR(150)	No	Lead Actor's first name
lead_actor _surname	VARCHAR(150)	Yes	Lead Actor's surname
lead_actor_age	INT	Yes	Lead Actor's current age
lead_actor_nationality	VARCHAR(100)	Yes	Lead Actor's nationality

Table 5: Actor Dimension Attributes.

Hierarchy:

lead_actor_nationality → lead_actor_surname → lead_actor_first_name

Attributes	Data Type	Nullable	Description
sk_director	INT	No	Surrogate key for director
bk_director	VARCHAR(150)	No	Business key for director
lead_director_first_name	VARCHAR(150)	No	Director's first name
lead_director_surname	VARCHAR(150)	Yes	Director's surname
lead_director_age	INT	Yes	Director's current age
lead_director_nationality	VARCHAR(100)	Yes	Director's nationality

Table 6: Director Dimension Attributes.

Hierarchy:

lead_director_nationality → lead_director_surname → lead_director_first_name

5.4.4. dim_production_company

The production company dimension captures organizational and geographical attributes of film production, including company name, city, country, and continent. This dimension enables geographical comparisons, identification of production hubs, and analysis of performance across production contexts.

Attributes	Data Type	Nullable	Description
sk_production_company	INT	No	Surrogate key for production company
bk_production_company	VARCHAR(150)	No	Business key for production company
production_company_name	VARCHAR(200)	No	Production company name
production_company_continent	VARCHAR(200)	Yes	Production company continent of origin
production_company_country	VARCHAR(200)	Yes	Production company country of origin
production_compay_city	VARCHAR(200)	Yes	Production company city of origin

Table 7: Production Company Dimension Attributes.

Hierarchy:

production_company_continent → production_company_country → production_company_city → production_company_name

5.5. Alignment with Business Needs

Every component of the Data Warehouse directly supports the business needs defined in section 2. The fact table provides the quantitative measures required for revenue and growth analysis, while the dimensions enable segmentation by time, content, talent, geography, and audience characteristics.

The decision to model a single fact table was intentional. Since all analytical questions relate to the same business process and grain, introducing additional fact tables would increase complexity without improving analytical capability. The resulting Data Warehouse is therefore both analytically complete and structurally efficient.

Dimensions	Key in Fact	Business Questions Supported
dim_date	Foreign key → dim_date.sk_date	Q7 – Genre Revenue Comparison; Q9 – Revenue by Age Classification; Q10 – Weekday Revenue Patterns; Q13 – Most Profitable Dates/Seasons.
dim_film	Foreign key → dim_film.sk_film	Q1 – Top Actors in Revenue; Q2 – Award vs Film Performance; Q3 – Performance vs Runtime Group; Q4 – Weekly Change by Genre; Q5 – Performance by Country; Q6 – Director Performance Overview; Q7 – Genre Revenue Comparison; Q8 – Revenue by Rating Bracket; Q9 – Revenue by Age Classification; Q11 – Top Languages in Revenue; Q12 – Top Films by Budget Group.

dim_production_company	Foreign key → dim_production_company. sk_production_company	Q5 – Performance by Country.
dim_director	Foreign key → dim_director.sk_director	Q6 – Director Performance Overview.
dim_actor	Foreign key → dim_actor.sk_actor	Q1 – top actors in revenue.

Table 8: Alignment of Dimensions with Business Questions.

This ensures a complete, cohesive, and business-driven dimensional model aligned with the informational needs of MAD Investment Group.

6. ETL - Implementation and Data Quality Controls

The ETL processes developed for MAD Investment Group were designed to transform heterogeneous and partially incomplete source data into a consistent, governed, and analytically reliable Data Warehouse. This section describes the ETL architecture adopted, the most relevant transformation decisions, and the automated data quality validation framework implemented to ensure that all data propagated to the analytical layer has undergone systematic validation and explicit review.

6.1. ETL Architecture and Execution Strategy

The ETL solution was implemented in Microsoft Fabric using structured, pipeline-based architecture. The overall flow is organized into three clearly separated layers: source ingestion, staging transformations, and Data Warehouse loading. This separation ensures a clean division of responsibilities and improves maintainability, transparency, and auditability.

A dedicated master pipeline was created to orchestrate the end-to-end ETL process, coordinating source ingestion, staging transformations, data quality validation, and final Data Warehouse.

Given the academic context of the project and the historical, non-volatile nature of the dataset, a full-load strategy was adopted. Each ETL execution processes the entire dataset end-to-end rather than relying on incremental or change-data-capture mechanisms. This approach simplifies validation, guarantees consistency across dimensions and facts, and ensures full reproducibility of results. Since the data does not change retroactively and no real-time requirements exist, a full-load strategy was deemed both sufficient and appropriate.

6.2. Staging Transformations: Key Design Decisions

All cleansing, standardization, and enrichment logic was intentionally concentrated in the staging area, ensuring that the Data Warehouse remains focused exclusively on analytical querying and performance.

Only non-trivial and analytically relevant transformations were implemented, including:

- Standardization of textual attributes (whitespace trimming, case normalization, capitalization).
- Explicit handling of missing and ambiguous values, preserving uncertainty where semantics are unclear.
- Construction of Business Keys for films, contributors, and production companies using SQL.
- Deduplication after Business Key assignment.
- Feature reduction for multi-valued attributes such as filming countries and spoken languages to control sparsity.
- Derivation of new attributes such as contributor age, name splitting, and structured awards metrics.
- Programmatic enrichment of the date dimension with holiday indicators, calendar seasons, and marketing seasons.

6.3. Dataset Splitting and Entity-Specific Source Preparation

After the initial ingestion of the source data, the ETL process introduced an explicit dataset-splitting step to support clearer entity-level processing and improve ETL maintainability. Although the consolidated dataset was analytically suitable, working with a single large file across all transformations would have increased pipeline complexity and reduced readability.

To address this, a first Dataflow was created to load the complete original dataset as a single entity. This Dataflow ("DF missing values and BK treatment") preserved the full structure of the source data while performing essential preparatory transformations. These included the treatment of missing and ambiguous values and the creation of all required Business Keys. At this stage, no entity separation was performed, ensuring that all subsequent transformations were applied consistently across the entire dataset.

The execution of this Dataflow produced a curated intermediate dataset, hereafter referred to as the *treated dataset*, which represents a standardized and governed version of the original source data. This treated dataset serves as the single authoritative input for all downstream Dataflows.

Building on this prepared dataset, five additional Dataflows were then developed to split the data into entity-specific source files. Each Dataflow extracted a subset of the data corresponding to a specific analytical entity, producing the following datasets:

- Film source data
- Actor source data
- Director source data
- Production company source data
- Daily box office source data

Importantly, during this splitting process, the selection of attributes was intentionally broader than the minimum required for dimension or fact tables creation. Rather than selecting only the final dimensional attributes, all columns that could plausibly hold analytical or governance relevance for each entity were retained. This design choice preserved flexibility in the ETL process, allowing later transformation, validation, and refinement decisions to be made without requiring re-extraction or reprocessing of the original data.

To orchestrate the execution of the Dataflows responsible for dataset preparation and entity separation, a dedicated pipeline was implemented. This pipeline ensures that global preparation steps, namely missing value handling and Business Key creation, are completed before any entity-specific Dataflows are executed.

Once the curated source dataset is available, the pipeline triggers the Dataflows that generate the entity-specific source files (film, actor, director, production company, and daily box-office) in parallel. This orchestration guarantees correct execution order, avoids dependency issues, and improves the robustness and reproducibility of the ETL process.

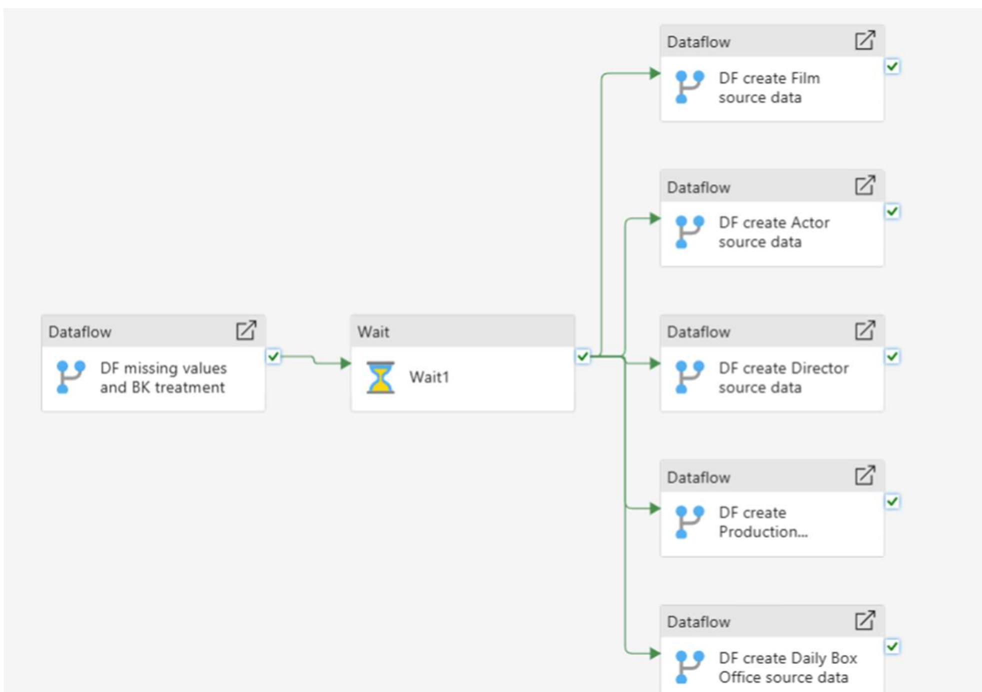


Figure 2: Split Source File Pipeline.

6.4. Business Key Construction Logic and Rationale

The construction of Business Keys (BKs) represents a central design decision in the ETL process, as these keys provide stable identifiers for analytical entities and support deduplication, referential integrity, and consistent dimensional joins. Because the source data did not provide reliable natural identifiers, all Business Keys were explicitly designed and implemented based on the semantic characteristics of the data and its intended analytical use.

Prior to Business Key creation, all contributing attributes were standardized to ensure consistency and reliability. This standardization included trimming leading and trailing whitespace, normalizing capitalization, and enforcing consistent character case across all relevant columns. These steps ensured that Business Key construction was not affected by superficial formatting differences and that logically identical entities were treated consistently.

6.4.1. Film Business Key

The Business Key for films was constructed using a combination of film title and production company name. An alternative approach using film title and release year was initially considered but subsequently rejected. Analysis of the data revealed that the same film may legitimately appear across multiple calendar years, for example when a film is released late in one year and continues its theatrical run into the following year. Using release year as part of the Business Key would therefore risk fragmenting a single film entity into multiple records.

The chosen design reflects the assumption that it is highly unlikely for a production company to release multiple distinct films with the same title. This approach minimizes ambiguity while remaining consistent with observed data behavior and the analytical grain of the model.

6.4.2. Production Company Business Key

The Business Key for production companies was defined using the combination of company name and city. This design supports differentiation between organizations with similar names operating in different locations, while avoiding over-reliance on country information.

6.4.3. Actor and Director Business Keys

The Business Keys for both actors and directors were constructed using the same logic: a combination of contributor name and birthdate. This approach provides a stable and sufficiently unique identifier for individual contributors, mitigating ambiguity in cases where different individuals share the same name.

6.4.4. Role of Business Keys in ETL and Data Quality Validation

All Business Keys defined in this stage were subsequently used as the foundation for deduplication logic and data quality validation rules. Ensuring Business Key uniqueness was a prerequisite for loading dimension tables and enforcing referential integrity between fact and dimension tables. By explicitly designing and documenting Business Keys, the ETL process establishes a clear, auditable, and semantically grounded foundation for the Data Warehouse.

6.5. ETL Investigation Note: Duplicate Treated Datasets

During the development of the staging and ETL processes, two treated datasets (treated_dataset and treat_dataset) were created as a result of an intermediate investigation step during Business Key implementation. During Business Key implementation, unexpected inconsistencies were identified, and a second treated dataset was created to isolate and investigate the issue without affecting the original workflow.

Further analysis showed that the issue was caused by an incorrect data type (TEXT/INT) in one of the attributes used for Business Key construction. Once corrected (TEXT), both treated datasets became fully equivalent in structure and content.

The duplication was only noticed after the workflow was already fully integrated in Microsoft Fabric. Both datasets have been used interchangeably and are functionally identical. Removing one at a late stage could introduce unintended effects in dependent pipelines, so both were retained. This decision does not impact data quality or analytical correctness and reflects a pragmatic approach to system stability close to the delivery deadline.

6.6. Staging Area Dataflows and Core Transformation

After Business Key construction, the ETL process continued with the development of a set of Dataflows responsible for loading the Staging Area tables. These Dataflows concentrate the majority of the transformation logic required to convert the curated source data into dimension- and fact-ready datasets, while keeping the Data Warehouse loading phase focused on structural integrity and key resolution.

The transformations implemented in the Dataflows go beyond basic technical adjustments and reflect deliberate analytical and modeling decisions.

6.6.1. Attribute Standardization and Structural Alignment

Within the Dataflows, attribute names were systematically renamed to follow consistent and meaningful naming conventions aligned with the dimensional model. Data types were explicitly enforced to ensure compatibility with downstream aggregation and analytical operations, particularly for dates, numeric measures, and categorical attributes.

These transformations establish a structurally consistent Staging Area that can be reliably consumed by the Data Warehouse loading pipelines.

6.6.2. Derivation of Analytical Attributes

Several analytically relevant attributes were derived within the Dataflows. For both actors and directors, first name and surname were extracted from the original full name fields, enabling clearer attribute semantics and supporting name-based hierarchies in the corresponding dimensions.

It was observed that some contributors are identified by a single name only, for which no reliable surname can be derived. In these cases, the surname attribute was intentionally left as NULL, and the corresponding column was modeled as nullable to preserve the original data semantics and avoid introducing artificial assumptions.

Contributor age was derived from birthdate information to support age-based analysis. This derivation was implemented with awareness of its time-dependent nature and is therefore treated as a descriptive attribute rather than a transactional measure.

6.6.3. Feature Reduction for Multi-Valued Attributes

The source dataset includes multi-valued attributes such as filming countries and spoken languages, originally represented across multiple expanded columns. Filming countries were initially spread across 14 columns, while spoken languages occupied 11 columns.

To avoid excessive sparsity and unnecessary model complexity, feature reduction was applied during the staging Dataflows. Filming countries were reduced from 14 to 3 attributes (primary, secondary, and tertiary), and spoken languages from 11 to 2 (primary and secondary).

By prioritizing the most relevant values, the model remains expressive enough to support geographical and linguistic analysis while avoiding excessive sparsity, simplifying downstream queries, and improving overall performance of the Data Warehouse.

6.6.4. Awards Attribute: Data Quality Challenge and ETL Resolution

During the staging phase, the Awards attribute was identified as a significant data quality and modeling challenge. In the source dataset, awards information was provided as unstructured free text, combining multiple distinct concepts (such as total award wins, awards nominations, Oscar wins, and Oscar nominations), within a single textual field. These values appeared in heterogeneous and inconsistent formats (e.g., *“9 wins & 22 nominations total”*, *“Won 1 Oscar, 73 wins & 152 nominations total”*, *“Nominated for 3 Oscars”*).

As a first step, value replacement transformations were applied to normalize key semantic terms and eliminate linguistic variation (e.g., *wins* → *win*; *nominations* → *nomination*; *Oscars* → *Oscar*). This ensured that equivalent concepts were represented consistently across all records and was a necessary prerequisite for reliable downstream parsing logic.

Due to the remaining heterogeneity of the award strings, a simple delimiter-based split was not feasible. Instead, custom columns were implemented within the Dataflows to apply rule-based parsing logic, extracting numeric values based on the standardized keywords (e.g., *win*, *nomination*, *Oscar*). These custom columns isolate each award-related concept independently and handle multiple textual patterns robustly.

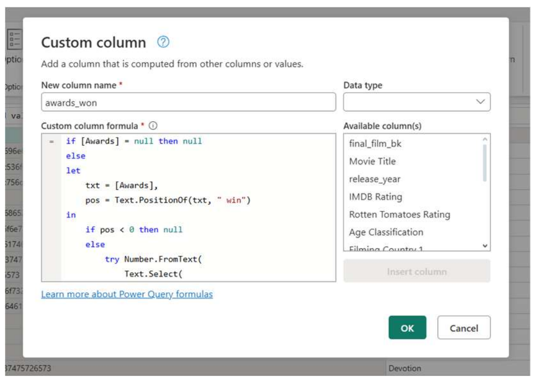


Figure 3: Custom Column Transformation for Award Parsing in the Film Dimension.

As a result, the original unstructured Awards field was transformed into a set of structured, numeric attributes, representing total awards won, awards nominations, Oscar won, and Oscar nominations. This transformation enables accurate aggregation and analytical use within the Film dimension, while preserving the semantic meaning of the source data and avoiding unsupported assumptions.

6.6.5. Genre Attribute: Semantic Ambiguity and ETL Resolution

During the staging phase, a data quality and semantic issue was identified in the genre attribute. For some films, multiple genres were provided within a single field, with up to three genres concatenated together. However, these genres were listed in alphabetical order rather than by narrative or commercial relevance, meaning that the first listed genre did not reliably represent the film's primary genre.

Because the ordering of genres carried no semantic meaning, deriving primary and secondary genres based on position would have introduced arbitrary assumptions and potential analytical distortion. To avoid this, we deliberately did not attempt to infer genre hierarchy.

Instead, a single consolidated genre attribute was created, preserving the original multi-genre combination as provided by the source. This approach maintains semantic fidelity to the source data while enabling consistent grouping and aggregation at the genre-combination level, without imposing unsupported assumptions about genre dominance.

6.6.6. Handling Non-Numeric Placeholders and Nullable Measures

In the source dataset, quantitative attributes such as *days_since_release* and *gross_revenue_per_day* occasionally appear as missing or as non-numeric placeholders (e.g., "–"), meaning the value is unavailable rather than equal to zero. Because fact table measures must be numeric, these placeholders were standardized during ETL by converting them to NULL, ensuring data type consistency while preserving the semantic intent of the source.

Both measures were therefore modeled as nullable in the staging layer (and retained as nullable in the fact load when applicable). This prevents indefinite values from being incorrectly interpreted as zeros, which could distort analytical outputs, especially in aggregations such as averages, trends, and lifecycle-based comparisons. Allowing NULLs also aligns with dimensional modeling best practices, since analytical functions naturally exclude NULL values, maintaining the correctness and reliability of derived insights.

```sql
CREATE TABLE fact_daily_box_office (
    fk_date                     INT                 NOT NULL,
    fk_film                     INT                 NOT NULL,
    fk_production_company       INT                 NOT NULL,
    fk_director                 INT                 NOT NULL,
    fk_actor                    INT                 NOT NULL,

    gross_revenue_per_day       DECIMAL (18,2)      NULL,
    days_since_release          INT                 NULL
);
```

Figure 4: Fact Table Definition with Nullable Measures to Preserve Missing Semantics.

6.6.7. Deduplication and Entity Consistency

Dataflows also implemented deduplication logic after Business Key assignment. This step ensured that each film, actor, director, and production company is represented only once in the Staging Area, preventing duplicated dimension members and safeguarding analytical correctness.

Deduplication was performed conservatively, preserving entity integrity and avoiding the accidental removal of legitimately distinct records.

6.6.8. Date Dimension Construction

Unlike other dimensions, the Date dimension was entirely constructed within the ETL process and does not originate from external source data. A dedicated Dataflow was developed to generate a complete and deterministic date dimension, including a system-generated surrogate key and a comprehensive set of calendar, weekday, holiday, and seasonal attributes.

Because the Date dimension is fully generated by design and does not rely on Business Keys or source-system values, its structure and contents are deterministic and consistent across executions. As a result, it does not require the same data quality validation rules applied to source-derived dimensions.

This dimension provides the temporal backbone for the Data Warehouse and supports time-based analyses such as weekday patterns, seasonal trends, and release-timing evaluation.

6.6.9. Pipeline-Orchestrated Loading of the Staging Area

To ensure a consistent and reliable population of the Staging Area, a dedicated pipeline was implemented to orchestrate the execution of Dataflows responsible for loading staging dimensions and facts. This pipeline enforces execution order, manages dependencies, and prevents partial or inconsistent loads.

The pipeline begins by clearing the existing staging tables using SQL statements, ensuring a clean state before each load. This full-refresh strategy was adopted to guarantee that each execution produces a clean and reproducible Staging Area state, avoiding residual data from previous runs. Once the Staging Area is cleaned, the pipeline triggers the Dataflows responsible for loading the staging dimension tables, including Date, Film, Actor, Director, and Production Company. These Dataflows are executed in parallel, as they are independent at the staging level.

After all staging dimensions have been successfully loaded, a second synchronization point ensures that dimensional data is fully available. Only then is the Dataflow responsible for loading the Daily Box Office staging fact executed. This sequencing guarantees referential consistency, as fact records depend on the presence of corresponding dimension keys.

By centralizing execution control in the pipeline, the ETL process ensures predictable behavior, simplifies monitoring and re-execution, and enforces correct load dependencies between staging components.

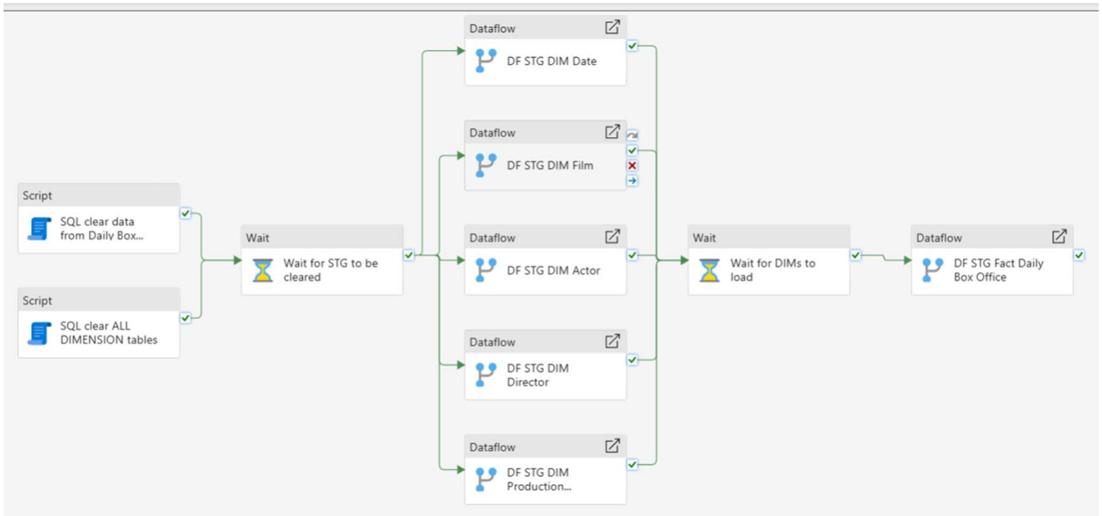


Figure 5: Pipeline for Loading Dimensions and Fact Table into the Data Staging Area.

6.7. Automated Data Quality Validation Framework

After loading the Staging Area tables, a dedicated data quality validation pipeline was executed before propagating data to the Data Warehouse. This validation framework was implemented as a centralized pipeline to ensure that all source-derived tables are fully validated prior to any downstream loading.

By centralizing rule execution and logging within a single orchestration layer, the pipeline guarantees a consistent validation order, prevents partial execution, and ensures that downstream processes only consume data that has passed the defined quality checks.

All validation results are recorded in a centralized quality log table, providing a transparent and auditable record of each rule execution. The objective of this framework is to detect structural, semantic, and referential issues early in the ETL process, thereby reducing the risk of propagating errors into analytical layers.

For clarity, the validation approach distinguishes between:

- **Technical data quality rules** (Rule #1–#4), which ensure structural correctness.
- **Business data quality rules** (B1–B3), which ensure semantic and analytical validity.

The Date dimension does not undergo the same data quality validation checks as other dimensions, as it is entirely system-generated during the ETL process and does not originate from external source data. Its structure and contents are deterministic by design, eliminating the need for Business Key, deduplication, or attribute-level validation.

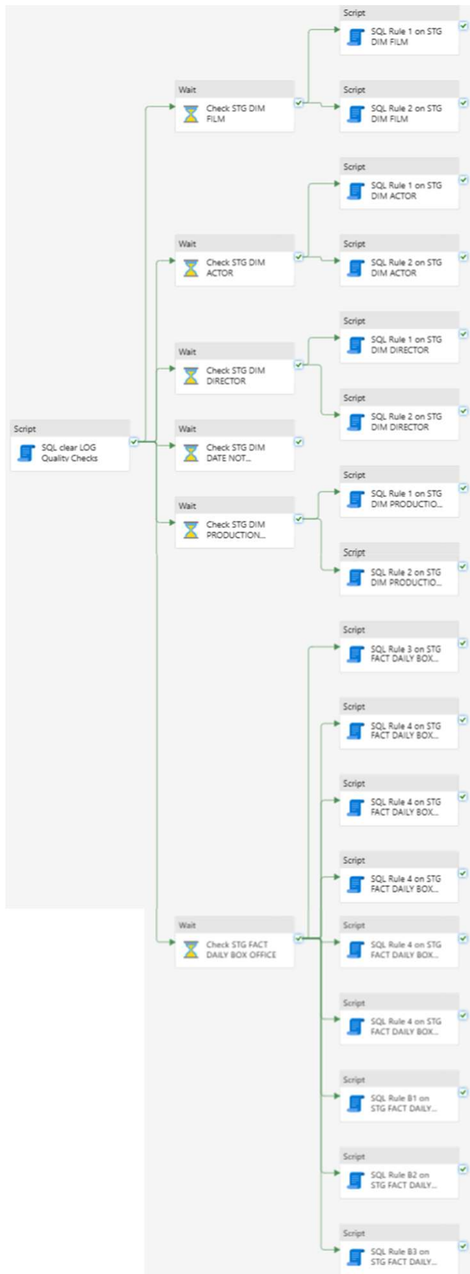


Figure 6: Pipeline for Quality Validation

6.8. Technical Data Quality Rules

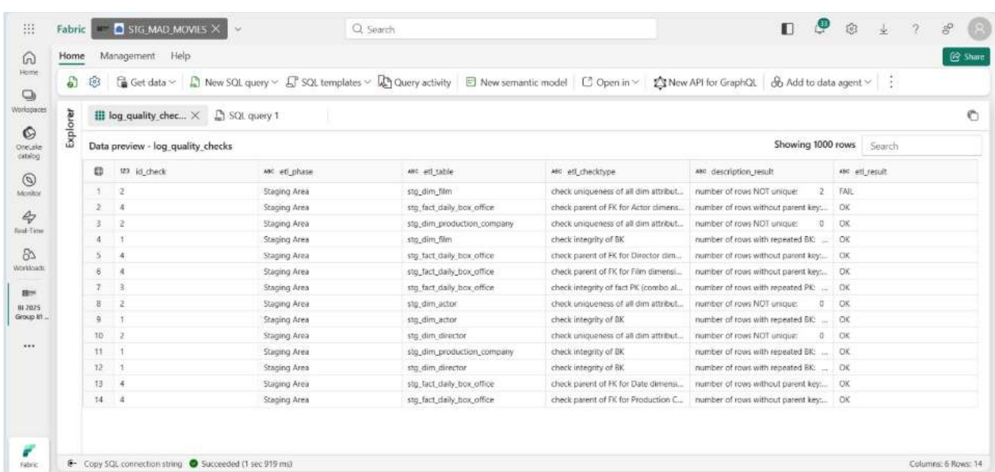
<u>Rule #1 – Business Key Integrity</u>

The first validation rule checked the integrity of Business Keys in all dimension tables. This rule verifies that each Business Key is unique and can effectively act as a stable surrogate identifier in the Data Warehouse. Ensuring Business Key uniqueness is critical to prevent duplicated dimension members. All dimension tables passed this validation, confirming that the deduplication logic applied during the Dataflow loads was effective.

<u>Rule #2 – Attribute-Level Uniqueness in Dimensions</u>

During the execution of the Staging Area data quality validation pipeline, a single failure was detected in the stg_dim_film table. This rule verifies whether each dimension row is uniquely identifiable based on a defined set of descriptive business attributes.

The validation log reported that two film records were not unique when evaluated using the following attributes: film title, release year, runtime, primary language, and main filming country. At first glance, this result appeared inconsistent, as a *Remove Duplicates* step had already been applied during the Dataflow that loads the Film dimension into the Staging Area.



Data preview - log_quality_checks

	id_check	etl_phase	etl_table	etl_checktype	description_result	etl_result
1	2	Staging Area	stg_dim_film	check uniqueness of all dim attribut...	number of rows NOT unique: 2	FAIL
2	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Actor dimens...	number of rows without parent key:...	OK
3	2	Staging Area	stg_dim_production_company	check uniqueness of all dim attribut...	number of rows NOT unique: 0	OK
4	1	Staging Area	stg_dim_film	check integrity of BK	number of rows with repeated BK: ...	OK
5	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Director dim...	number of rows without parent key:...	OK
6	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Film dimensi...	number of rows without parent key:...	OK
7	3	Staging Area	stg_fact_daily_box_office	check integrity of fact PK (combo al...	number of rows with repeated PK: ...	OK
8	2	Staging Area	stg_dim_actor	check uniqueness of all dim attribut...	number of rows NOT unique: 0	OK
9	1	Staging Area	stg_dim_actor	check integrity of BK	number of rows with repeated BK: ...	OK
10	2	Staging Area	stg_dim_director	check uniqueness of all dim attribut...	number of rows NOT unique: 0	OK
11	1	Staging Area	stg_dim_production_company	check integrity of BK	number of rows with repeated BK: ...	OK
12	1	Staging Area	stg_dim_director	check integrity of BK	number of rows with repeated BK: ...	OK
13	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Date dimensi...	number of rows without parent key:...	OK
14	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Production C...	number of rows without parent key:...	OK

Figure 7: Staging Area Data Quality Validation Results Logged via ETL Pipeline.

Further analysis on this failure is addressed later in the 6.8.1. sub-section.

Rule #3 – Fact Table Grain Validation

For the fact table, a grain validation rule was applied to ensure that each row was uniquely identified by the combination of all its Foreign Keys. This check confirms that the fact table does not contain duplicated events at the same dimensional intersection (e.g. the same film, date, actor, director, and production company). Maintaining a well-defined and enforced grain is essential to guarantee the correctness of analytical measures. The fact table passed this validation successfully.

Rule #4 – Referential Integrity Between Fact and Dimensions

Finally, referential integrity checks were executed to validate that every Foreign Key present in the fact table corresponds to an existing Business Key in the related dimension tables. These checks ensure that no orphan fact records exist, which could otherwise result in incomplete or inconsistent analytical results. All Foreign Key relationships were validated successfully, confirming the consistency between fact and dimension data in the Staging Area.

6.8.1. Film Dimension Validation Failure (Rule #2): Investigation and Resolution

To understand the cause of this failure, targeted SQL queries were executed in the Staging Area. The analysis confirmed that the affected records corresponded to the films ***Shaun of the Dead*** and ***Spider-Man: No Way Home***, each appearing twice with identical descriptive attributes. Several attributes, such as film runtime, were missing (NULL), increasing the likelihood of collisions in the attribute-level uniqueness check.

Messages Results

	bk_...	film_title	film_rele...	film_genre	film_age_...	primary_l...	secondar...	film_runti...	imdb_rati...	rt_rating	film_bud...	main_fil...	secondar...	
1	536861756e2...	Shaun of the ...	2024	ComedyHorror	R	English	NULL	NULL	NULL	NULL	NULL	United Kingd...	France	Un
2	536861756e2...	Shaun of the ...	2024	ComedyHorror	R	English	NULL	NULL	NULL	NULL	NULL	United Kingd...	France	Un
3	53706964657...	Spider-Man: ...	2022	ActionAdvent...	PG-13	English	Tagalog	NULL	NULL	NULL	NULL	United States	NULL	NU
4	53706964657...	Spider-Man: ...	2022	ActionAdvent...	PG-13	English	Tagalog	NULL	NULL	NULL	NULL	United States	NULL	NU

Figure 8: Film Dimension Rows Violating Attribute Uniqueness Rule.

An examination of the Film Business Key (bk_film) revealed that these rows had different Business Key values, which explains why the deduplication step, applied only on the Business Key, did not remove them, in line with Rule #1.

Messages Results

	film_title	bk_film	main_filming_country	bk_len
1	Shaun of the Dead	536861756e206f662074686520446561642c3230323420...	United Kingdom	94
2	Shaun of the Dead	536861756e206f662074686520446561642c526f6775652...	United Kingdom	64
3	Spider-Man: No Way Home	5370696465722d4d616e3a204e6f2057617920486f6d65...	United States	124
4	Spider-Man: No Way Home	5370696465722d4d616e3a204e6f2057617920486f6d65...	United States	94

Figure 9: Film Records Sharing Descriptive Attributes but Differing in Business Key.

Further joins with the Fact and Production Company dimensions showed that each duplicated film record was associated with a different production company (e.g. original release versus re-release by a different distributor). Because the Film Business Key was deliberately constructed using the combination of film title and production company, each row represents a distinct business entity under the defined dimensional grain.

Messages Results

	bk_film	film_title	film_release_year	primary_language_film	main_filming_country	production_company_name
1	536861756e206f66207468652044 6...	Shaun of the Dead	2024	English	United Kingdom	2024 Re-releaseFocus Features
2	536861756e206f66207468652044 6...	Shaun of the Dead	2024	English	United Kingdom	Rogue Pictures
3	5370696465722d4d616e3a204e6f2...	Spider-Man: No Way Home	2022	English	United States	2022 Re-releaseSony Pictures Relea...
4	5370696465722d4d616e3a204e6f2...	Spider-Man: No Way Home	2022	English	United States	Sony Pictures Releasing

Figure 10: Film Records Differentiated by Production Company Despite Identical Descriptive Attributes.

Rule 2 failure therefore reflected a misalignment between the validation rule and the dimension grain of the Film dimension, rather than a true data quality issue.

To resolve this, the uniqueness validation rule for the Film dimension was updated to include the Business Key. After re-execution, all validation checks passed successfully, confirming the readiness of the Staging Area for loading into the Data Warehouse.

6.9. Business Data Quality Rules

In addition to technical validation, three business data quality rules were implemented for the daily box office fact table to ensure semantic and analytical consistency.

<u>Business Rule B1 – Gross Revenue Must Be Non-Negative</u>

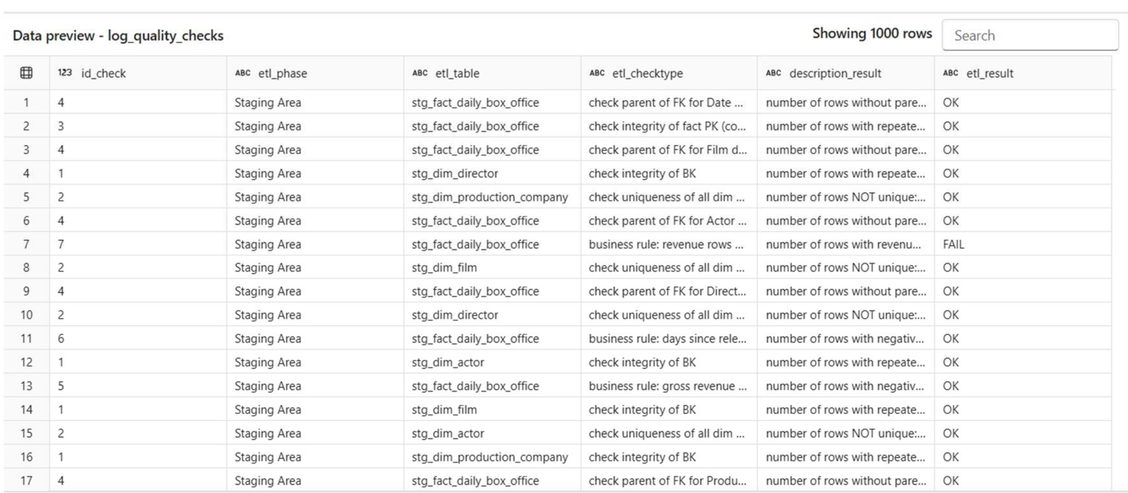
This rule enforces that, when populated, gross_revenue_per_day is always greater than or equal to zero, as negative box office revenue has no business meaning and would indicate data extraction or processing errors. The rule passed successfully, confirming the validity of all revenue values.

<u>Business Rule B2 – Days Since Release Must Be Non-Negative</u>

This rule ensures that, when populated, days_since_release contains only non-negative values. Negative values would be analytically meaningless and compromise lifecycle-based analysis. The rule passed successfully for all applicable records.

<u>Business Rule B3 – Revenue Rows Must Have Days Since Release</u>

This rule enforces that whenever gross_revenue_per_day is populated, days_since_release should also be available to support lifecycle-based analysis. The rule identified a small number of exceptions, which were investigated further and are discussed in the following sub-section (6.9.1).



Data preview - log_quality_checks Showing 1000 rows Search

	id_check	etl_phase	etl_table	etl_checktype	description_result	etl_result
1	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Date ...	number of rows without pare...	OK
2	3	Staging Area	stg_fact_daily_box_office	check integrity of fact PK (co...	number of rows with repeate...	OK
3	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Film d...	number of rows without pare...	OK
4	1	Staging Area	stg_dim_director	check integrity of BK	number of rows with repeate...	OK
5	2	Staging Area	stg_dim_production_company	check uniqueness of all dim ...	number of rows NOT unique:...	OK
6	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Actor ...	number of rows without pare...	OK
7	7	Staging Area	stg_fact_daily_box_office	business rule: revenue rows ...	number of rows with revenu...	FAIL
8	2	Staging Area	stg_dim_film	check uniqueness of all dim ...	number of rows NOT unique:...	OK
9	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Direct...	number of rows without pare...	OK
10	2	Staging Area	stg_dim_director	check uniqueness of all dim ...	number of rows NOT unique:...	OK
11	6	Staging Area	stg_fact_daily_box_office	business rule: days since rele...	number of rows with negativ...	OK
12	1	Staging Area	stg_dim_actor	check integrity of BK	number of rows with repeate...	OK
13	5	Staging Area	stg_fact_daily_box_office	business rule: gross revenue ...	number of rows with negativ...	OK
14	1	Staging Area	stg_dim_film	check integrity of BK	number of rows with repeate...	OK
15	2	Staging Area	stg_dim_actor	check uniqueness of all dim ...	number of rows NOT unique:...	OK
16	1	Staging Area	stg_dim_production_company	check integrity of BK	number of rows with repeate...	OK
17	4	Staging Area	stg_fact_daily_box_office	check parent of FK for Produ...	number of rows without pare...	OK

Figure 11: Results of Automated Data Quality Validation in the Staging Area.

6.9.1. Business Rule B3: Failure Analysis and Justification

The investigation revealed two distinct scenarios:

- For ***Bones and All***, the first failing record corresponded to the actual release date (18/11/2022), where a value of days_since_release = 0 would be appropriate. However, subsequent days (19/11 and 20/11) also lacked this value, indicating limitations in the source data.
- For ***Psycho***, ***Rear Window***, and ***Vertigo***, the failures occurred for films originally released in the 20th century. In these cases, the box office records correspond to modern re-releases, making it impossible to accurately derive days_since_release without reliable original release date information.

These failures were not caused by incorrect revenue data, but by limitations in the availability and meaning of lifecycle information for catalogue titles and re-releases.

	film_title	fk_date	gross_revenue_per_day	days_since_release
1	Bones and All	2022-11-18	57887.00	NULL
2	Bones and All	2022-11-19	35526.00	NULL
3	Bones and All	2022-11-20	27591.00	NULL
4	Psycho	2024-12-27	181230.00	NULL
5	Rear Window	2024-12-27	858030.00	NULL
6	Vertigo	2024-12-27	158085.00	NULL

Succeeded (1 sec 745 ms) Copilot completions: Off Columns: 4 Rows: 6

Figure 12: Fact Records with Revenue and Missing Days Since Release (Business Rule B3 Exceptions).

Following this analysis, the identified exceptions were deliberately accepted while keeping Business Rule B3 in place. Enforcing the rule strictly would require deriving values that cannot be reliably justified from the available data. Rather than introducing artificial assumptions or removing valid revenue records, the decision was to preserve the factual data and explicitly document the analytical limitations.

Overall, the rule fulfilled its purpose by highlighting records where lifecycle-based analysis may not be fully supported. Given that the vast majority of records comply with the rule, the presence of a small number of well-understood exceptions was considered acceptable within the scope of this project.

6.10. Data Quality Limitations

During the analysis of production company attributes, a data quality limitation related to geographical completeness was identified. In several records, continent and city information were available, while the corresponding country attribute was missing. Although in some cases the country could be inferred from the city, such values were intentionally left as NULL. This decision was made to preserve fidelity to the source data and avoid introducing external assumptions or enrichment logic beyond what is explicitly provided. The presence of this inconsistency was documented as a semantic completeness issue rather than corrected, ensuring transparency and traceability in downstream analysis.

6.11. Data Warehouse Loading Pipeline and Key Management

After completion of the Staging Area loads and the execution of the data quality validation pipeline, a dedicated Data Warehouse loading pipeline was executed to populate the dimensional model. This pipeline was implemented using Copy Data activities, enabling controlled loading from the Staging Area tables into the Data Warehouse tables.

6.11.1. Full-Load Strategy and Table Clearing

Given the historical and non-volatile nature of the dataset, the Data Warehouse loading was designed as a full-load process. To guarantee repeatability and avoid duplicated records across executions, the pipeline begins by clearing previously loaded data in the Data Warehouse.

This approach ensures that each execution produces a consistent Data Warehouse state aligned with the validated staging outputs.

6.11.2. Surrogate Key Generation for Dimensions

To support efficient joins and ensure stable dimensional modeling practices, surrogate keys (SKs) were introduced in the Data Warehouse for all dimensions except the Date dimension. The Date dimension already contains an “intelligent” surrogate key based on the calendar date, making additional generated keys unnecessary.

For the remaining dimensions (Film, Actor, Director, and Production Company), surrogate keys were generated during loading. This guarantees deterministic key assignment within each full-load execution and establishes an internal, warehouse-level identifier independent from the source system representation.

Conceptually, this step converts the Staging Area’s Business Keys (used for traceability and governance) into Data Warehouse surrogate identifiers (used for performance and dimensional integrity).

6.11.3. Fact Loading with Foreign Key Lookup

The Staging Area fact table stores foreign keys as Business Key values, consistent with the staging design. However, the Data Warehouse fact table must reference dimension surrogate keys to ensure efficient star-schema joins.

For this reason, the fact loading step performs a lookup transformation during the Copy Data activity: each Business Key stored in the staging fact table is joined to its corresponding dimension row in the Data Warehouse, and the correct surrogate key is retrieved. The resulting output contains the translated foreign keys alongside the measures and is then inserted into the Data Warehouse fact table.

This lookup-based translation ensures referential integrity between the fact table and all dimensions, consistent star schema relationships and improved query performance, since joins occur on integer surrogate keys rather than textual Business Keys.

6.11.4. Pipeline Orchestration

The pipeline enforces the correct loading order required by dimensional modeling: Data Warehouse tables are cleared (full-load reset); all dimensions are loaded first (Film, Actor, Director, Production Company, Date); only after dimension load completion is the fact table loaded (Daily Box Office).

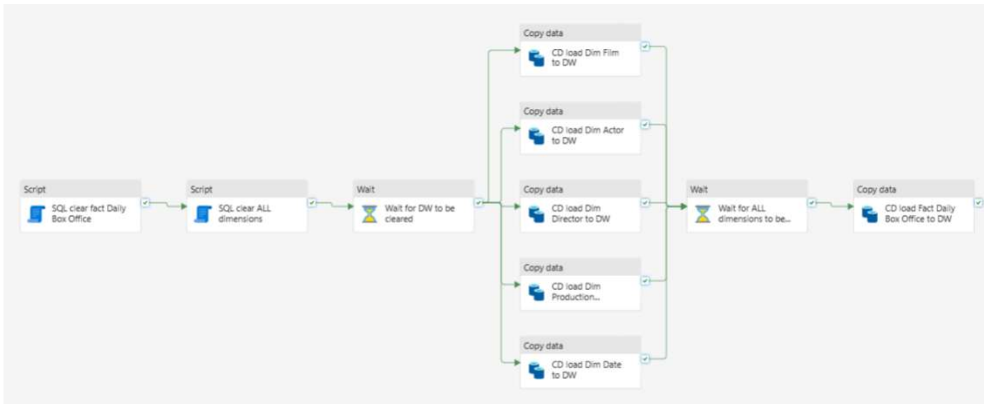


Figure 13 - Data Warehouse Loading Pipeline Orchestration

This ordering ensures that all surrogate keys exist before the fact load lookup step is executed, preventing orphan references and guaranteeing a consistent analytical model.

6.12. Master ETL Pipeline Orchestration (Original ETL Work)

In addition to the individual pipelines developed for source preparation, staging, validation, and Data Warehouse loading, a master ETL pipeline was implemented to orchestrate the complete end-to-end data integration process. This pipeline acts as the single execution entry point for the entire ETL workflow, coordinating all subordinate pipelines in a controlled and repeatable sequence.

6.12.1. Rationale for a Centralized Master Pipeline

The primary role of the master pipeline is to coordinate the execution of all major ETL phases (source preparation, staging transformations, data quality validation, and Data Warehouse loading) in a deterministic and repeatable manner.

Without a centralized master pipeline, each ETL component would need to be executed independently, increasing operational complexity and the risk of incorrect execution order. Such fragmentation could lead to situations where downstream processes consume incomplete or unvalidated data. The master pipeline addresses this risk by enforcing a strict and explicit execution sequence, ensuring that each processing stage completes successfully before the next one begins.

By centralizing orchestration logic, the solution improves reliability, reduces manual intervention, and guarantees consistency across ETL runs. Each execution represents a complete and traceable ETL cycle, from raw source data to the final analytical layer.

6.12.2. Orchestration Strategy

The master pipeline invokes each major ETL phase as a dedicated sub-pipeline, using Invoke Pipeline and Wait activities to enforce strict execution order. The orchestration follows a clear logical sequence:

1. Source data preparation and splitting;
2. Initial staging area loading;
3. Data quality validation;
4. Final Data Warehouse loading.

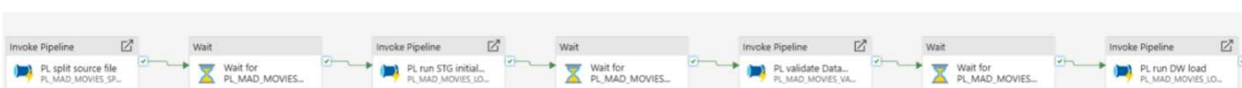


Figure 14 - ETL Master Pipeline Orchestration.

Wait activities are used to guarantee that each phase completes successfully before the next phase begins, preventing downstream processes from consuming incomplete or unvalidated data.

7. Critical Review / Lessons Learned

Considering that none of the group members had a prior background in Business Intelligence, this project represented our first independent practical experience with the development of a complete BI solution. While the concepts were introduced during the classes, it was through the hands-on use of Microsoft Fabric that the subject truly became clear. Implementing the pipelines, staging logic, validations, and dimensional model helped us understand the purpose of each step and how the different components of a BI architecture fit together.

One of the main lessons learned was how to transform data in a clean and structured way while keeping business meaning in mind. Working with real box-office data showed that missing and ambiguous values are common, particularly in cases such as film re-releases or incomplete lifecycle information. Applying Kimball's principles helped us understand when uncertainty should be preserved instead of forcing artificial assumptions.

The project also reinforced the importance of early modeling decisions. Defining the grain and Business Keys upfront proved essential to building a consistent data mart capable of answering business questions related to film performance, contributors, and release timing. Several validation cases showed that what initially appeared to be data quality issues often reflected legitimate business realities.

Overall, this project demonstrated that Business Intelligence is not only about technical implementation, but about aligning data, business context, and analytical goals. In the context of MAD Investment Group, the project showed how a well-designed BI solution can support more informed and data-driven film investment decisions.

8. Conclusions

The Business Intelligence solution developed for MAD Investment Group transforms historical box-office data into a structured and reliable analytical platform that supports evidence-based film investment decisions. By integrating box-office performance with curated contextual attributes related to films, talent, time, geography, and audience reception, the project provides a comprehensive view of the factors that influence cinematic success.

The resulting Data Warehouse enables MAD Investment Group to answer its core business questions, including identifying profitable genres, optimal release periods, high-performing actors and directors, and the relationship between ratings, awards, and financial performance. These capabilities reduce uncertainty in decision-making and allow creative intuition to be complemented with empirical evidence.

From a technical perspective, the implementation of a clean dimensional model, supported by a robust ETL process and automated data quality validation, ensures that analytical results are based on consistent, governed, and auditable data. This increases confidence in the insights produced and allows stakeholders to focus on strategic interpretation rather than data reliability.

9. Webgraphy:

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10. Appendix

Data Dictionary: Original Metadata

Movie Title – The name of the film.

Production Studio – The company that produced the movie.

Date – The specific calendar day of the recorded data.

Gross Revenue per Day – How much money the film made on that day.

Theaters – Number of cinemas showing the movie on that day.

Average – Average revenue per theater for that day.

Gross Change by Day – The change in revenue compared to the previous day.

Gross Change by Week – The change in revenue compared to the previous week.

Total Gross Revenue – The total amount of money the film has made so far.

Days Since Release – The number of days that have passed since the movie's premiere.

Lead Director – The main director of the film.

Lead Actor – The main actor starring in the film.

Lead Director Birth Year – The year the director was born.

Lead Actor Birth Year – The year the lead actor was born.

TMDB Rating – Viewer rating from The Movie Database.

IMDB Rating – Viewer rating from IMDb.

Rotten Tomatoes Rating – The percentage of positive ratings given by audience members on Rotten Tomatoes, reflecting how regular viewers received the film.

Age Classification – The film's age restriction (e.g., PG, R, 16+).

Filming Country – The country where the movie was filmed.

Spoken Language – The main language spoken in the film.

Genre – The film's category (e.g., Action, Comedy, Drama).

Runtime (min) – Length of the movie in minutes.

Production Company Country – The country where the production company is based.

Director Nationality – The country where the lead Director was born

Actor Nationality - The country where the lead Actor was born

Budget – The movie's budget

Poster URL – The link of the picture of the respective movie poster

Franchise – If the film has a sequel

Awards – How many awards the movie was nominated for and/or won (overall and Oscars)

Production Company City - The city where the production company is based

Production Company Continent - The continent where the production company is based